24 May 1965
SUPERSEDING
MIL-I-23208
28 February 1962

MILITARY SPECIFICATION

INDICATORS, SIGHT, LIQUID, REFRIGERANT

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers liquid refrigerant sight flow indicators, for systems using refrigerant 12 or 22.

* 1.2 Classification.- Indicators shall be of the following types and sizes as specified (see 6.2):

Type I	-	External flare connection end.
Type II	-	Solder joint connection end.
Size	-	3/8-inch connections.
Size	-	1/2-inch connections.
Size	-	5/8-inch connections.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

BB-C-310	-	Chlorofluoro Hydrocarbons (of the Methane and Ethane Series).
UU-T-111	-	Tape; Paper, Gummed (Sealing and Securing).
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-636	-	Box, Fiberboard.
PPP-B-665	-	Boxes; Paperboard, Metal-Stayed (including Stay Material).
PPP-B-676	-	Boxes, Set-Up, Paperboard.

MILITARY

MIL-P-116	-	Preservation, Methods of.

/FSC 6680/

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STANDARDS

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MS 17798 - Indicator, Sight, Liquid Refrigerant.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNDERWRITERS' LABORATORIES, INC. (UL)

No. 207(c) - Refrigerant-Containing Components

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago 11, Illinois; Walt Whitman Road, Melville, L.I., New York, or 1655 Scott Boulevard, Santa Clara, California.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

* 3.1 Preproduction sample.- When specified (see 6.2), before production is commenced, a sample indicator covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for inspection and approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.2 Standard product.- The indicators delivered under this specification shall be manufacturer's standard commercial production except for any design modifications and additional requirements specified herein.

3.3 Safety standard.- Indicators shall be tested and approved under UL Standard No. 207(c).

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3.3.1 Certification.- Prior to approval of the preproduction sample, if one is submitted, or prior to approval of the first shipment of indicators, the supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the indicators he proposes to furnish under this specification meet the requirements of UL Standard No. 207(c). Acceptable evidence of meeting the requirements of this standard shall be the UL label, an approved UL listing, or a certified test report acceptable to the Government, indicating that the indicators offered have been tested and conform to the requirements of UL Standard No. 207(c).

3.4 Interchangeability of parts.- All parts bearing the same manufacturer's part number shall be completely interchangeable without modification.

☆ 3.5 Materials.- The materials shall conform to the applicable specifications, standards, and requirements specified herein. Materials not definitely specified shall be of the quality used in the manufacturer's standard indicators.

☆ 3.6 Design.- Indicators shall be designed to allow an unrestricted flow of dichlorodifluoromethane, CCL_2F_2, or monochlorodifluoromethane, $CHClF_2$, refrigerants conforming to type 12 and 22, respectively, of BB-C-310, while allowing an observation of the flow through an incorporated sight glass or glasses. Unless otherwise specified (see 6.2), the indicator shall incorporate moisture indicating features.

☆ 3.7 Construction.- Overall dimensions of the body, flare fittings and solder joints shall conform to MS 17798. The sight glass shall be protected from dirt and foreign matter with a cap or cover if manufacturer's design requires.

3.7.1 Sight glass.- Unless otherwise specified (see 6.2), sight glass shall be of the single or double port type at the manufacturer's option. Each sight glass shall contain not less than 0.240 square inch of unrestricted view area. When the sight glass is of the replaceable glass type, it shall be provided with gaskets of a material to prevent leakage.

3.7.2 Moisture indicating feature.- When required, (see 3.6), the moisture indicating feature shall be of the visual type and shall indicate whether the refrigerant is wet or dry when tested as specified in 4.4.1. A color comparison code or chart, or a printed code or chart for refrigerant type 12 and type 22 of BB-C-310 shall be incorporated into or permanently attached to the indicator. The indicating feature shall provide determination of a wet or dry refrigerant, and caution determination if manufacturer's design requires.

3.8 Finish.- The indicators shall have the manufacturer's standard commercial finish. The finished indicators shall not have any sharp burrs or edges.

3.9 Workmanship.- The finished indicators shall conform to the quality and grade of product established by this specification.

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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Preproduction sample inspection.- When a preproduction sample is required, it shall be examined for the defects listed in table I and for dimensions specified and tested as specified in 4.4.1.

4.3 Sampling for inspection and acceptance.- Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

* 4.3.2 End product inspection.- For the purpose of sampling, the inspection lot shall consist of all the indicators of one type and size offered for inspection at one time. The sample unit for the inspection shall be one indicator.

* 4.3.2.1 Visual examination.- Examination shall be made of the end product for the defects in table I. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 defects, expressed in terms of defects per hundred units.

TABLE I.- Examination of visual defects

Examine	Defect
Finish	Sharp burrs or edges.
Design, construction, and workmanship general (applicable to all components and assemblies)	Part missing. Parts not properly joined or assembled.
Color code or chart (when applicable)	Missing, not legible, not affixed in the prescribed manner.

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* 4.3.2.2 Dimensional examination.- Examination shall be made of the end item to determine compliance with dimensional requirements. Any dimension that does not meet specified requirements shall be classified a defect. The inspection level shall be S-2 and the AQL shall be 4.0, expressed in terms of defects per hundred units.

* 4.3.2.3 Testing of the end product.- The test specified in 4.4.1 shall be performed using an inspection level of S-4 and an AQL of 4.0, expressed in terms of defects per hundred units.

4.3.3 Code and standard compliance.- Proof of compliance with UL Standard No. 207(c) requirements of 3.3 shall be made available to the Government representative.

* 4.3.4 Examination of preparation for delivery.- An examination shall be made to determine that packaging, packing, and marking as required by section 5 of this specification are complied with. Defects shall be scored as specified in table II. The sample unit shall be one shipping container prepared for delivery, except that the container shall not be sealed. The lot shall be the number of containers offered for delivery at one time. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units. Examination for closure and strapping shall be in accordance with the applicable box specification.

TABLE II.- Preparation for delivery defects

Examine	Defect
Markings (exterior and interior container)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials (exterior and interior container)	Component missing, damaged, or otherwise defective.
Contents (exterior and interior container)	Number more or less than required.

4.4 Tests.-

4.4.1 Moisture element test.- The moisture indicating element shall be tested to determine compliance with 3.7.2. Refrigerant type 12 and type 22 of BB-C-310 shall be used for the test. The refrigerant shall be passed through the indicator at 90°F. ± 3°F. Each refrigerant shall have a maximum of 10 parts per million (ppm) moisture content when passed through and the indicating element shall indicate dry in accordance with the manufacturer's established code or chart.

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The moisture content shall be gradually increased to a maximum of 40 ppm. The element shall show a definite change to a wet condition. An intermediate change in the element shall be observed if manufacturer's design requires. The manufacturer's standard method for determining the moisture content of the refrigerant shall be acceptable.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.-

* 5.1.1.1 Unit packaging.- The two connection ends of each indicator shall be sealed with plastic, metal or wood caps or plugs. Each indicator shall be packaged in a snug-fitting, folding, set-up or metal stayed paperboard box conforming to style II, type A, class a, or style III or VIII, type B, class a of PPP-B-566; type I, grade A or B, class A, style 3 or 4 of PPP-B-676; or style C of PPP-B-665, respectively. Closure of folding and set-up paperboard boxes shall be in accordance with the appendix of the applicable box specification. Metal stayed paperboard boxes shall be closed with minimum 2-inch width gummed paper tape conforming to class 2 of UU-T-111. When specified (see 6.2), each indicator shall be packaged method IC-2 of MIL-P-116.

* 5.1.1.2 Intermediate packaging.- Twenty-four type I or type II indicators, unit packaged as specified in 5.1.1.1, shall be packaged in a snug-fitting fiberboard box conforming to style RSC, type CF or SF, class domestic, variety SW, grade 125 of PPP-B-636. Closure shall be in accordance with the appendix of the box specification. When indicators are unit packaged method IC-2 (see 5.1.1.1), the intermediate box shall conform to style RSC, W6c or W6s of PPP-B-636.

5.1.2 Level C.- Indicators shall be packaged in accordance with the industry's practice.

5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

* 5.2.1 Level A.- One hundred and forty-four type I or ninety-six type II indicators, packaged as specified in 5.1, shall be packed in a snug-fitting, fiberboard box conforming to style RSC, V3c or V3s of PPP-B-636, except that when weight of contents exceeds 65 pounds, the style and grade shall be FTC and V3s, respectively, and the weight limitation shall be waived. Each shipping container shall be closed, waterproofed, and reinforced in accordance with the appendix of the container specification. When weather-resistant intermediate boxes are used (see 5.1.1.2), waterproofing of the shipping container shall not be required.

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* 5.2.2 Level B.- Unless otherwise specified (see 5.2.2.1), one hundred and forty-four type I or ninety-six type II indicators, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF or SF, class domestic, grade 275, variety SW of PPP-B-636, except that when weight of contents exceeds 65 pounds, the style, type, grade and variety shall be FTC, CF, 350 and DW, respectively. Closure shall be in accordance with the appendix of the container specification. Style FTC containers shall be reinforced as specified in 5.2.1.

5.2.2.1 When specified (see 6.2), weather-resistant fiberboard specified in 5.2.1 shall be used for the shipping container.

5.2.3 Level C.- Indicators, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The indicators covered by this specification are intended to be inserted in a liquid refrigerant line to permit observation of the refrigerant flow and when provided with a moisture indicating feature, to determine whether the refrigerant is wet or dry.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and size required (see 1.2).
(c) When preproduction sample is required (see 3.1).
(d) When moisture indicating feature is not required (see 3.6).
(e) Sight glass if other than specified (see 3.7.1).
(f) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(g) When waterproof unit packaging is required (see 5.1.1.1).
(h) When weather-resistant fiberboard containers are required (see 5.2.2.1).

6.3 The outside margins of this document have been marked to indicate where changes from the previous issue have been made. An asterisk (*) indicates a substantive change in the paragraph text. The sign (#) indicates the addition of a completely new paragraph. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations.

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Custodians:

Army - GL
Navy - YD
Air Force - 11

Preparing activity:

Army - GL

Project No. 6680-0039-1

Review activities:

Navy - YD
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User activities:

Army - MO
Navy - MC, SH

Review and User information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing Of DOD Documents.

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"to detach this form, cut along this line"

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION: Indicators, Sight, Liquid, Refrigerant - MIL-I-23208A

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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